



AI-DRIVEN EV CHARGING MANAGEMENT

UrbanEV Intelligence Framework

An end-to-end AI-agent system for EV charging demand forecasting, dynamic tariff optimization, and continuous network monitoring — built on **2.13 million real-world charging observations** across 247 urban stations.

Project at a Glance

This project develops an integrated AI-agent framework for EV charging management using the UrbanEV dataset — one of the most comprehensive urban charging records available. Three autonomous agents work in sequence to forecast demand, optimize pricing, and monitor network health in real time.

247

Charging Stations

Urban network coverage

8,640

Time Intervals

Temporal granularity

2.13M

Observations

Total records analyzed

28.0%

Avg Utilization

Network-wide average

568

Critical Alerts

High-risk periods flagged

Why It Matters: With only 28% average utilization, network expansion may not be immediately necessary — the priority is redistributing demand through intelligent forecasting and dynamic pricing.

Data Landscape & Preprocessing

Data Pipeline

Raw Inputs	Purpose
volume.csv	Charging demand
occupancy.csv	Charger usage
duration.csv	Session duration
information.csv	Station metadata
time.csv	Temporal attributes

Engineered Features

- timestamp
- utilization_rate
- occupancy_density
- revenue
- queue_proxy

Final Dataset

Stations	247
Time Steps	8,640
Records	2,134,080
Features	15+

- ✓ No missing values
- ✓ No duplicates
- ✓ Complete temporal coverage

Key EDA Findings & Demand Behavior Insights

Low utilization

Average network utilization is **28.0%**, leaving major spare capacity.

Uneven usage

A few stations absorb most demand; many remain underused.

Dynamic pricing wins

Priced stations outperform on volume, revenue, and occupancy density.

Non-CBD leads

Non-CBD stations deliver **2.5×** more volume and **2.1×** more revenue than CBD.

📌 **Bottom line:** Demand is localized — targeted actions beat broad expansion.

Demand Prediction Modeling & Results

A **Random Forest Regressor** was trained on 2.13 million observations across 247 stations, using 15+ engineered features including occupancy, duration, charger counts, utilization rate, and temporal variables to forecast charging demand volume.

Model Performance

MAE	4.25
RMSE	13.77
R ²	0.984

An R² of 0.984 confirms the model captures nearly all variance in charging demand, supporting reliable operational planning.

Top Demand Drivers

- 01

Fast Charger Count
- 02

Session Duration
- 03

Slow Charger Count
- 04

Occupancy
- 05

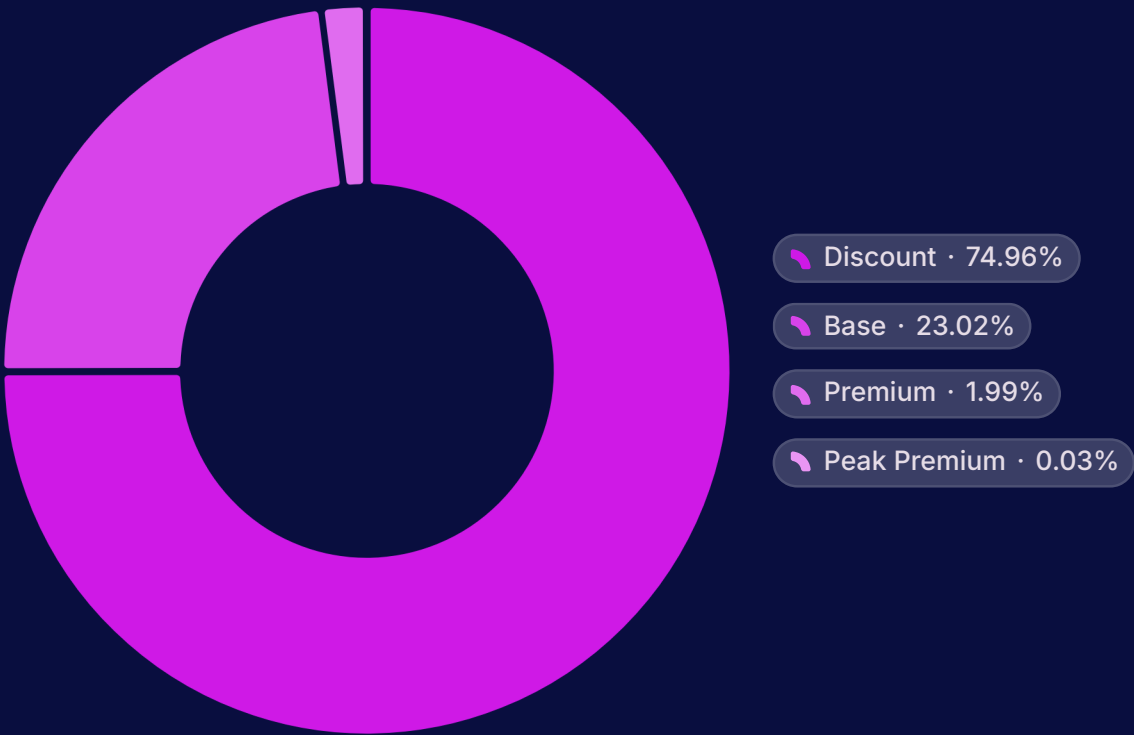
Hour of Day

Infrastructure availability and session characteristics are the strongest demand determinants.

Dynamic Tariff Optimization Logic & Pricing Outcomes

A rule-based demand scoring system maps real-time conditions to four tariff tiers — from Discount to Peak Premium — activating higher prices only during genuine congestion. The framework responds dynamically to occupancy, queue proxy, and utilization signals.

Tariff Distribution



Tariff Framework

Demand Score	Tariff Applied
Low	Discount
Medium	Base
High	Premium
Very High	Peak Premium

Key Insight: Only 0.03% of intervals trigger Peak Premium pricing, confirming the framework targets genuine congestion — not routine demand.

Monitoring Agent & Feedback Loop Performance

The Monitoring Agent continuously evaluates Utilization Rate, Occupancy Density, and Demand Score across all 247 stations. It identified **568 critical alerts** — concentrated in a small subset of station-time combinations — enabling targeted operational response.

Network Summary

Average Utilization	28.0%
Average Demand Score	18.6%
Critical Alerts	568



Operational risk is highly concentrated:

only 0.03% of intervals triggered Peak Premium pricing, yet 568 critical alerts were identified.

Agent Feedback Loop

01	02
Demand Prediction	Tariff Optimization
03	04
Monitoring	Alert Generation
05	
Operational Action	

Business, Operational & Policy Implications

Results align with existing EV charging literature, which suggests demand is highly concentrated across stations and peak periods — making forecasting and dynamic pricing valuable operational tools. The framework delivers actionable value for three key stakeholder groups.

For Operators

- Improve charger utilization through demand redistribution
- Prioritize maintenance and staffing at high-demand stations
- Reduce congestion using dynamic pricing signals

For Customers

- Transparent, demand-responsive pricing
- Better charger availability via predictive load balancing
- Reduced wait times during peak periods

For Infrastructure Planning

- Focus capital investments on high-demand corridors
- Delay expansion in persistently underutilized locations
- Use demand forecasts to justify site selection

Assumptions & Limitations

This appendix clarifies the modeling choices behind the analysis and the boundaries of interpretation. Together, these assumptions and constraints define what the results can reliably support today — and where future improvements could add the most value.

Key Assumptions

The framework assumes historical UrbanEV patterns remain useful for near-term inference and forecasting.

- Demand patterns are persistent
- Observed demand reflects user behavior
- Tariff multipliers are static and rule-based
- Operational context is relatively stable
- Non-modeled factors are treated as exogenous
- Single-city findings can still be diagnostic

Project Limitations

The analysis is constrained by the available data and modeling approach.

- Single-city UrbanEV dataset
- No real-time deployment pipeline
- No electricity wholesale price integration
- No weather or seasonal variables
- Deterministic pricing logic
- Limited causal interpretation



Next steps: real-time deployment, weather-aware modeling, wholesale price integration, and reinforcement learning-based pricing.

Project Outcome

This project delivers an end-to-end EV charging intelligence framework demonstrating that an integrated AI-agent architecture can support scalable, data-driven charging management through forecasting, dynamic pricing, and continuous monitoring.



Forecast Demand

Random Forest model with $R^2 = 0.984$ on 2.13M observations



Optimize Pricing

Four-tier dynamic tariff responding to real-time demand signals



Monitor Network

568 critical alerts identified across 247 stations



Drive Decisions

Python · Pandas · Scikit-Learn · Time-Series Feature Engineering

An integrated AI-agent framework can support scalable EV charging management through forecasting, dynamic pricing, and continuous monitoring — with results consistent with established EV charging literature.